Stream

Documentation Infrastructure Audit Report

Prepared by VeriOps

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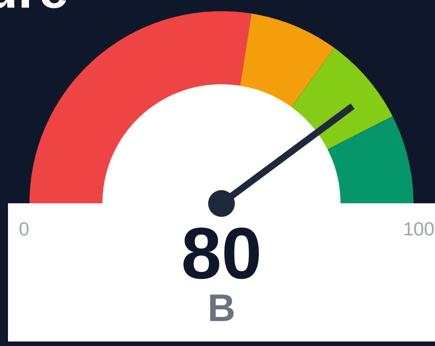


Risk: Moderate

Key Metrics

Pages scanned: 27093

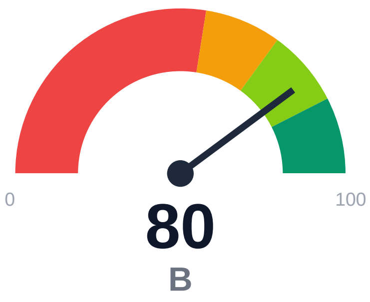
Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Crawl coverage is low: 69.56% (40000 URLs examined of 57506 in discovered scope). Run authenticated mode or in
- Broken internal links: 903
- Link verification inconclusive for 213 URLs (auth/rate-limit/server block); these are excluded from broken-link counts.
- SEO/GEO issue rate is high: 25.24%

Executive Summary



Audit covered 27,093 pages across 4 documentation sites (Chat, Video, Activity Feeds, Moderation) with 69.56% crawl coverage due to 40,000 URL examination limit against 57,506 discovered pages. 903 confirmed broken internal documentation links exist across all properties. API reference coverage stands at 75.0% with 16,573 API pages detected. Only 25.92% of pages display last-updated metadata, and 25.24% of pages exhibit SEO/geo issues. Overall confidence rating is 79.3%.

External posture: SEO/GEO issue rate **25.2%**, public API coverage **75.0%**, broken links **903**.

80 Audit Score	**B** Grade	**Moderate** Risk Band
27093 Pages Scanned	**903** Broken Links	**25.2%** SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://getstream.io/activity-feeds/docs	6520	243	80.0	99.4	24.6
https://getstream.io/moderation/docs	7000	214	70.0	99.5	25.8
https://getstream.io/video/docs	6478	244	80.0	99.4	24.5
https://getstream.io/chat/docs	7095	202	70.0	99.5	26.0

Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	**92+**	**-12**
Industry average (developer docs)	**85**	**-5**
Minimum acceptable	**70**	**+10**
Your score	**80**	**-**

Gap to industry average: **5 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

API Coverage (Public)
75.0%

SEO/GEO Issue Rate (lower is better)
25.2%

Example Reliability (estimate)
99.4%

Freshness Visibility (last-updated metadata)
25.9%

Internal Metrics (from scorecard)

API Coverage (Internal)
0.0%

Example Reliability (Internal)
100.0%

Docs-Contract Drift
0.0%

Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
$140,822 annual exposure	**$249,513** annual exposure	**$431,728** annual exposure
Remediation: **$2,090** Monthly loss: **$4,125** Opportunity: **$7,436**/mo	Remediation: **$5,445** Monthly loss: **$12,903** Opportunity: **$7,436**/mo	Remediation: **$8,800** Monthly loss: **$27,808** Opportunity: **$7,436**/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	$1,782	$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	$1,122	$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	$2,257	$990
F-API-COVERAGE	Undocumented API operations	LOW	$2,904	$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	$2,244	$935
F-DRIFT	Code/docs contract drift	LOW	$1,901	$660
F-TERMINOLOGY	Terminology inconsistency	LOW	$693	$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix all 903 confirmed broken internal documentation links across Chat, Video, Activity Feeds, and Moderation docs [impact: critical, effort: high]

2. Increase crawl examination limit and run authenticated crawl to cover remaining 30.44% of discoverable pages (17,413 pages) [impact: critical, effort: medium]

3. Add last-updated metadata to 74.08% of pages currently missing freshness indicators [impact: high, effort: medium]

4. Resolve 25.24% SEO/geo issues affecting search visibility and international access [impact: high, effort: medium]

5. Close 25% API reference coverage gap and investigate 10,422 crawl trap URLs for IA improvements [impact: medium, effort: high]

Expert Analysis: Strengths and Risks

Strengths

+ High example code reliability at 99.44% across detected samples
+ Strong API page detection with 16,573 pages identified and 75% reference coverage confirmed
+ Multi-project topology correctly identified; no fragmentation penalty applied to federated documentation structure
+ Zero broken repository navigation links (GitHub/GitLab); all 903 broken links are user-facing documentation links
+ Crawl examined 40,000 URLs and successfully indexed 27,093 pages across 4 distinct product documentation sites

Risks

- 903 confirmed broken internal documentation links directly impact user navigation and trust
- 30.44% of discoverable content not crawled (17,413 pages) due to examination cap, hiding potential additional issues
- 25.24% SEO/geo issue rate threatens search discoverability and international user access
- 74.08% of pages lack last-updated timestamps, preventing users from assessing content currency
- 213 links unverified due to authentication/rate-limiting; actual broken link count may be higher
- 25% API reference gap leaves significant endpoint documentation potentially missing or undetected
- 10,422 URLs skipped as crawl traps indicate potential information architecture or URL generation problems

Limitations

* External audit cannot verify content accuracy, technical correctness, or whether documented APIs match actual implementation behavior
* 213 links remain unverified due to authentication requirements or rate limiting; actual broken link count may differ
* 30.44% of discovered pages not examined due to crawl cap; hidden issues in uncrawled content cannot be assessed
* Crawl trap detection (10,422 URLs skipped) may have excluded legitimate pages if URL patterns triggered false positives
* API coverage detection relies on page structure patterns; non-standard API documentation formats may be undercounted or missed

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.